



Anomaly Detection for Fraud Risk in Banking Data





Banking Transaction

City

All

2020

2021

2022

2023

2024

2025

Total Amount

PY: \$2.32M

\$3.09M

33.19% ▲ vs 2022



Total Tracsaction

PY: 7.79K

10.44K

33.98% ▲ vs 2022



Total Account Balance

PY: \$39.88M

\$53.22M

33.45% ▲ vs 2022



Avg Trans Value

PY: \$297.62

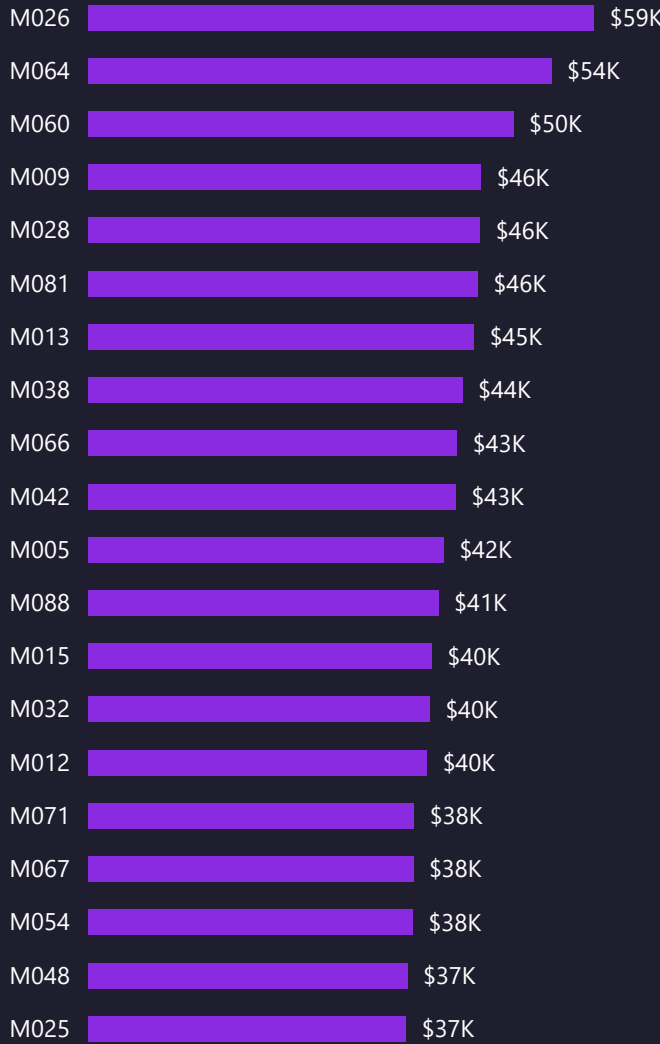
\$295.87

-0.59% ▼ vs 2022



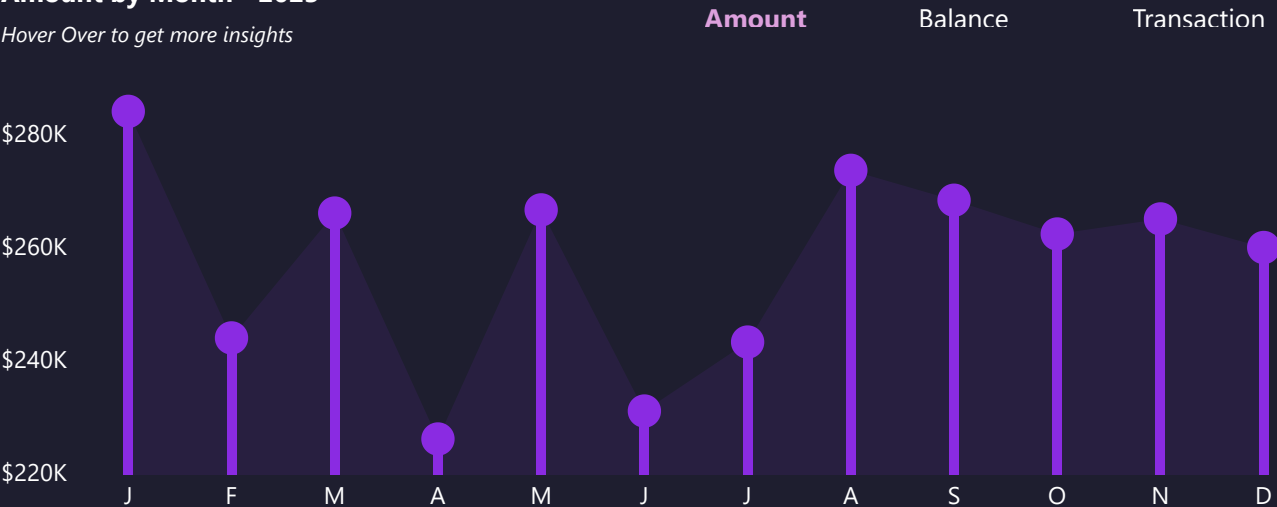
Top 20 Amount by Merchant - 2023

Hover Over to get more insights



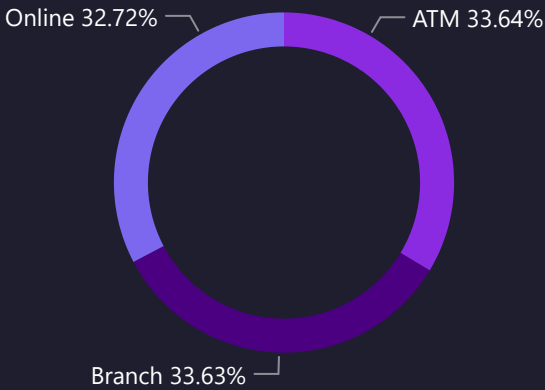
Amount by Month - 2023

Hover Over to get more insights



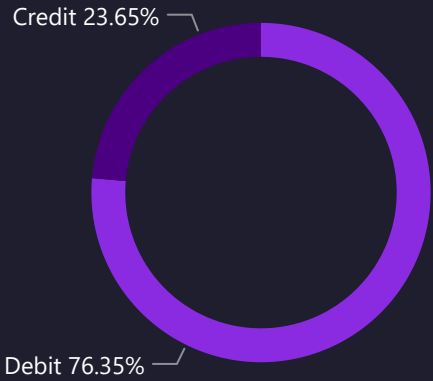
Amount by Channel - 2023

Hover Over to get more insights



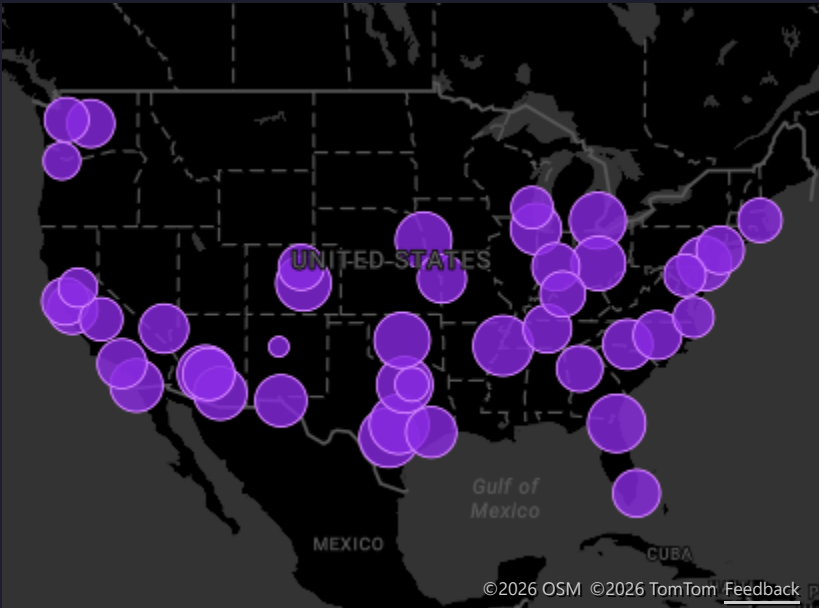
Amount by Transaction Type - 2023

Hover Over to get more insights



Amount by Location - 2023

Hover Over to get more insights



City [Top 5]	Amount %	Total Amount
Memphis	20.73%	\$94,616.00
San Antonio	20.46%	\$93,401.00
Austin	20.35%	\$92,897.00
Jacksonville	19.57%	\$89,312.00
Detroit	18.89%	\$86,232.00



Banking Transaction

MTD

QTD

YTD

City

All

2020

2021

2022

2023

2024

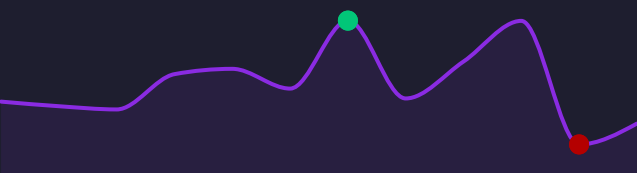
2025

Amount MTD

PY: \$259.74K

\$191.07K

-26.44% ▼ vs 2023



Transaction MTD by Channel

PY: 877

633

-27.82% ▼ vs 2023

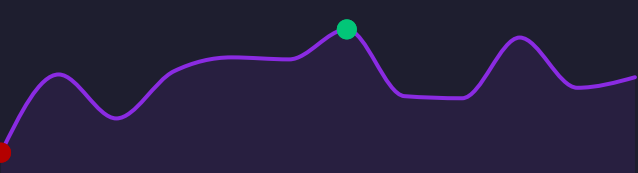


Avg Trans Value MTD

PY: \$296.17

\$301.84

1.92% ▲ vs 2023

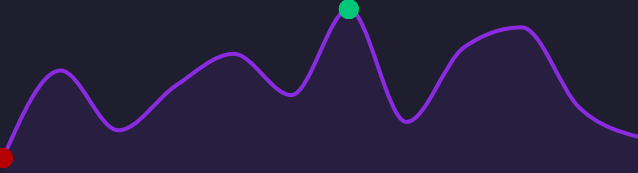


Avg Cus Value MTD

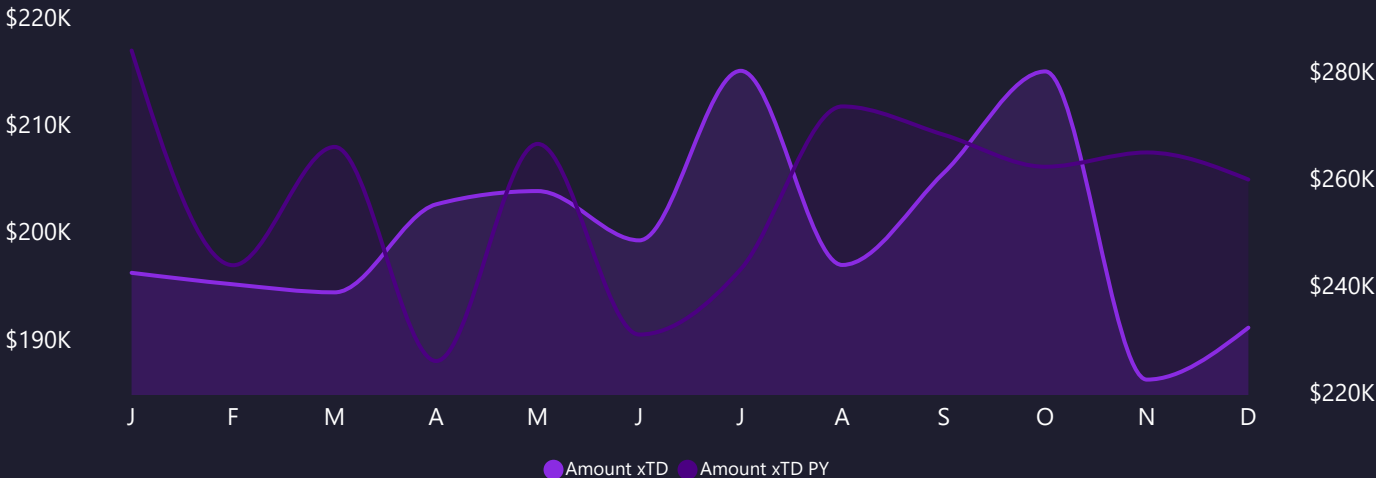
PY: \$664.29

\$557.04

-16.15% ▼ vs 2023



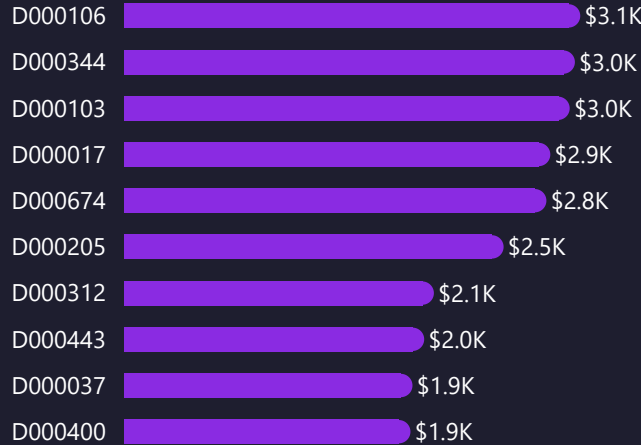
Amount MTD and Amount MTD PY by Month



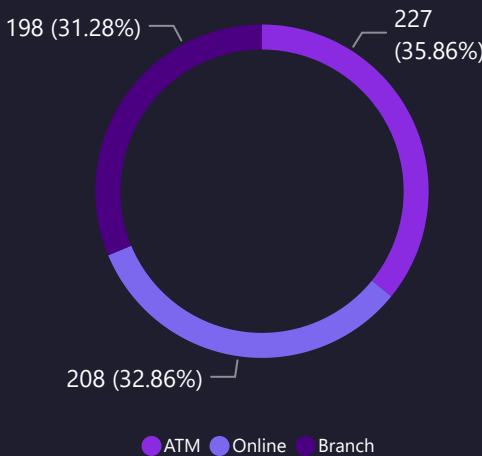
Top 20 ACV MTD by Customer



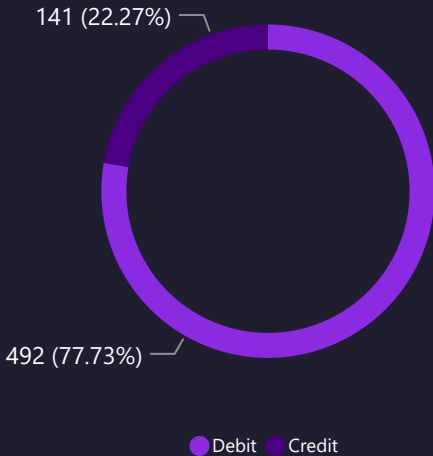
Top 10 Amount MTD by Device



Transaction MTD by Channel



Transaction MTD by Transaction Type



City [Top 7]	Trans Growth % xTD
Sacramento	-2.13%
Raleigh	-3.70%
San Diego	-6.67%
Oklahoma City	-10.94%
Columbus	-12.90%
El Paso	-15.69%
Indianapolis	0.00%



Banking Transaction

Amount

Balance

Transaction

City

All

2020

2021

2022

2023

2024

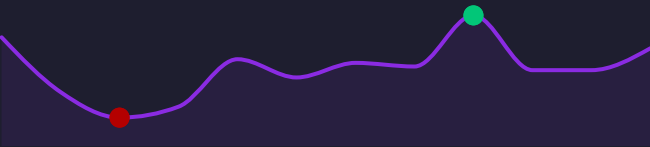
2025

Total Customer

PY: 494

0.00% ▼ vs 2021

494



R

Average Recency

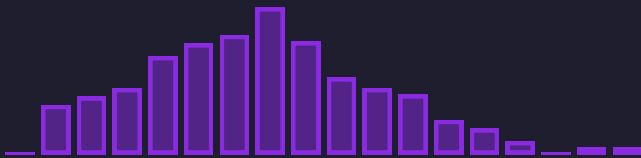
29



F

Average Frequency

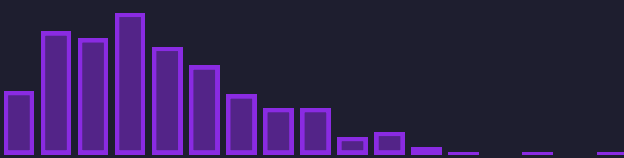
101



M

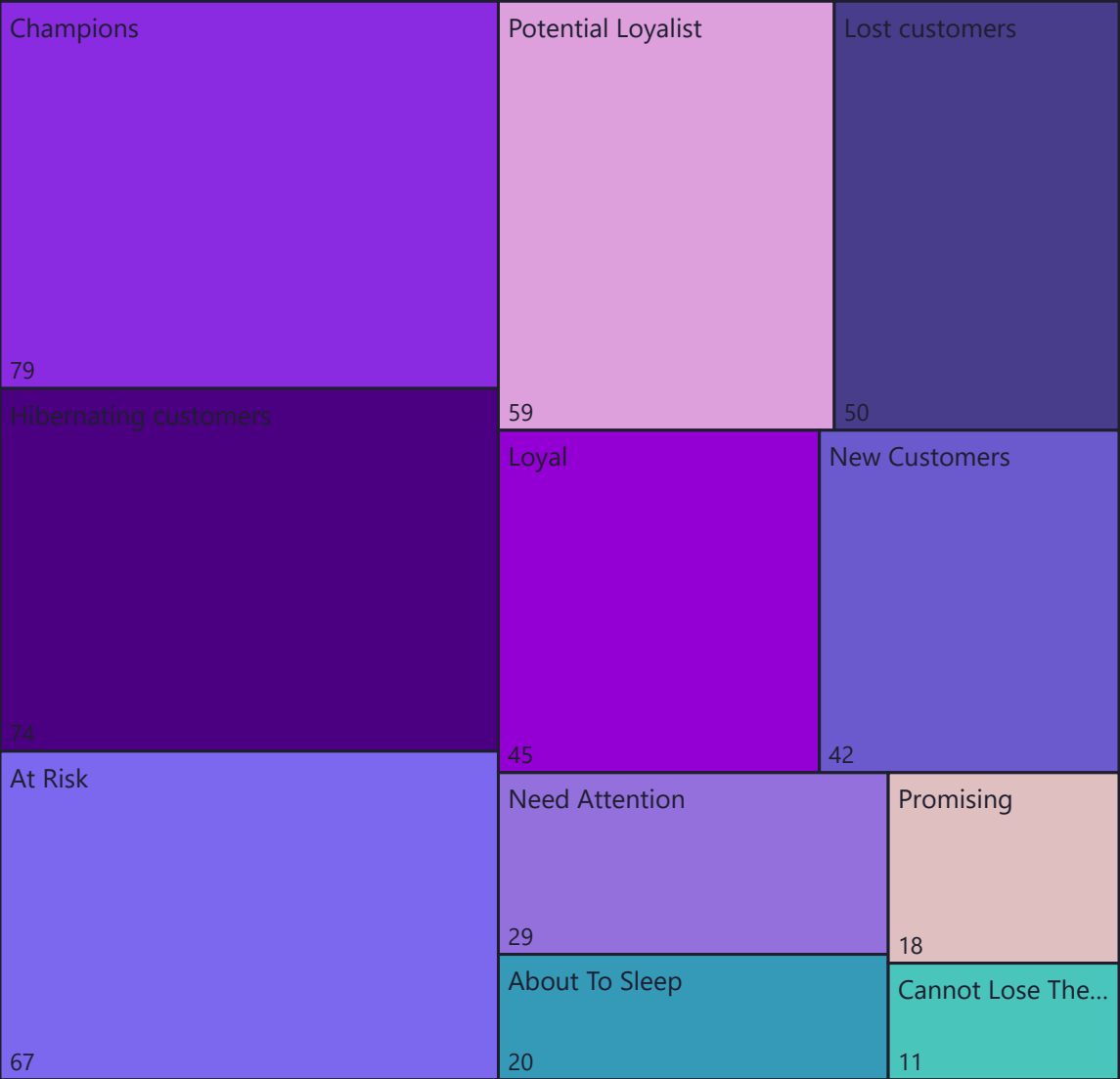
Average Monetary

\$30.09K



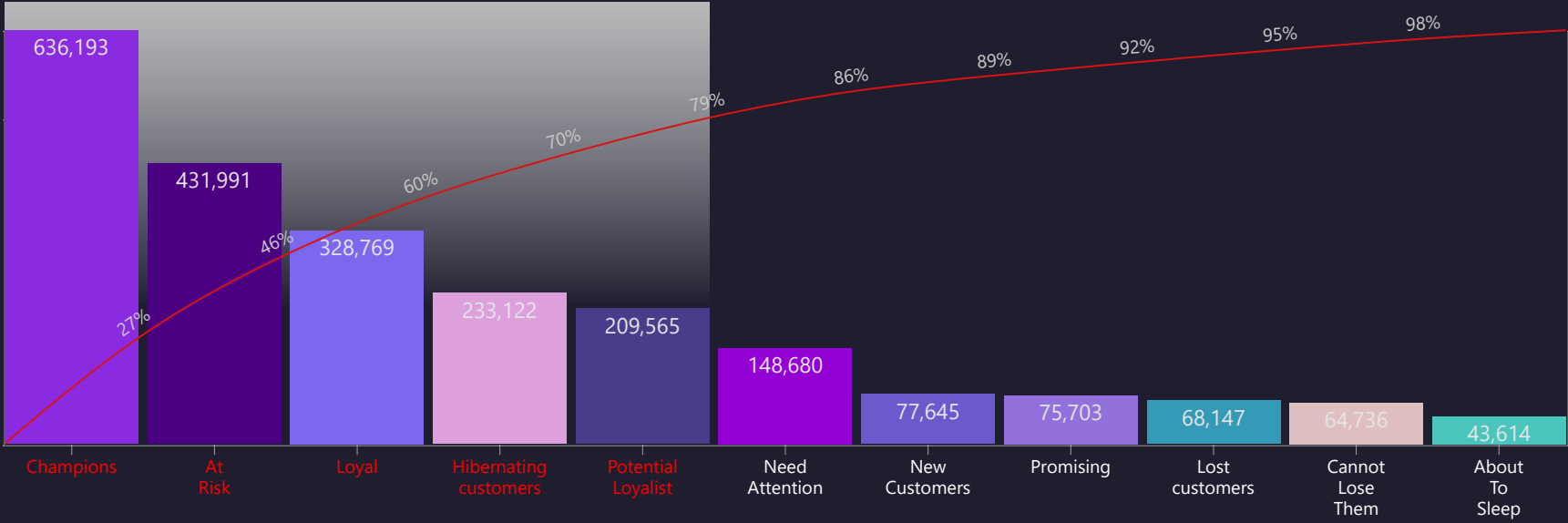
Total Customer by Segment Customer RFM - 2022

Hover Over to get more insights



Amount by Pareto Analysis - 2022

Hover Over to get more insights



AccountID	Segment Customer	RFM_Score	Recency	Frequency	Monetary
AC00460	Champions	555	2	238	\$111,657
AC00363	Champions	455	8	249	\$103,321
AC00337	At Risk	255	28	155	\$84,217
AC00095	Champions	555	3	179	\$81,857
AC00021	Champions	455	7	153	\$79,100
AC00020	Loyal	355	18	177	\$77,781
AC00136	Champions	555	1	159	\$77,692
AC000358	Loyal	355	21	156	\$76,172



Banking Transaction



Retention Customer by Month after First Transaction Month

First Transaction Month	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38
01/01/2020	350	262	261	265	264	255	248	269	259	270	261	279	264	235	266	261	261	253	254	269	260	273	254	265	269	250	249	254	261	257	266	265	267	263	259	266	300	277	287
02/01/2020	79	50	48	53	47	49	49	43	42	52	57	51	44	45	52	46	52	53	61	51	54	59	47	51	49	44	47	46	49	48	49	52	47	54	49	62	50	55	57
03/01/2020	33	20	20	20	14	21	17	21	17	20	18	16	20	15	17	17	18	19	16	21	12	14	15	19	19	19	18	22	14	18	24	17	15	19	24	21	19	20	20
04/01/2020	14	7	6	10	7	7	5	9	3	9	5	8	7	4	7	7	4	7	3	8	8	6	8	5	4	7	4	6	3	3	5	3	7	8	7	9	5	7	9
05/01/2020	7	2	5	3	3	4	4	1	2	3	3	4	3	3	2	4	4	3	1	4	5	2	4	2	4	1	4	3	3	3	4	2	4	3	4	2	2	4	1
06/01/2020	4	3	2	3	1		2	4	2	1	3	3	2	3	2	3	1	2	1	2	3	3	2	2	2	1	1	3	2	3	1	3	4	2	2	3	2	2	3
07/01/2020	3		1	1			1		2	1	1	2	1	3	2	1	2	1		1	1	1	1	2	1	1	2	2	1	2	2	3		1	2	2		1	2
11/01/2020	1													1		1					1				1			1				1		1	1				
12/01/2020	2		1	1	1	1	1	1				1				1	1	2	1	1			1			1	1		1	2	1	1	1	1	1	1	2		
02/01/2021	1	1										1	1			1															1			1		1			
04/01/2021	1				1							1		1			1	1								1			1				1		1	1			
Total	495	345	344	356	338	337	327	348	327	356	348	366	342	310	348	342	344	341	337	357	344	358	332	346	349	324	327	337	334	337	353	346	346	353	349	369	381	366	379

Retention Customer by Month after First Transaction Month

First Transaction Month	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
01/01/2020	100.00%	74.86%	74.57%	75.71%	75.43%	72.86%	70.86%	76.86%	74.00%	77.14%	74.57%	79.71%	75.43%	67.14%	76.00%	74.57%	74.57%	72.29%	72.57%	76.86%	74.29%	78.00%	72.57%
02/01/2020	100.00%	63.29%	60.76%	67.09%	59.49%	62.03%	62.03%	54.43%	53.16%	65.82%	72.15%	64.56%	55.70%	56.96%	65.82%	58.23%	65.82%	67.09%	77.22%	64.56%	68.35%	74.68%	59.49%
03/01/2020	100.00%	60.61%	60.61%	60.61%	42.42%	63.64%	51.52%	63.64%	51.52%	60.61%	54.55%	48.48%	60.61%	45.45%	51.52%	51.52%	54.55%	57.58%	48.48%	63.64%	36.36%	42.42%	45.45%
04/01/2020	100.00%	50.00%	42.86%	71.43%	50.00%	50.00%	35.71%	64.29%	21.43%	64.29%	35.71%	57.14%	50.00%	28.57%	50.00%	50.00%	28.57%	50.00%	21.43%	57.14%	57.14%	42.86%	57.14%
05/01/2020	100.00%	28.57%	71.43%	42.86%	42.86%	57.14%	57.14%	14.29%	28.57%	42.86%	42.86%	57.14%	42.86%	42.86%	28.57%	57.14%	57.14%	42.86%	14.29%	57.14%	71.43%	28.57%	57.14%
06/01/2020	100.00%	75.00%	50.00%	75.00%	25.00%		50.00%	100.00%	50.00%	25.00%	75.00%	75.00%	50.00%	75.00%	50.00%	75.00%	25.00%	50.00%	25.00%	50.00%	75.00%	75.00%	50.00%
07/01/2020	100.00%		33.33%	33.33%			33.33%		66.67%	33.33%	33.33%	66.67%	33.33%	100.00%	66.67%	33.33%	66.67%	33.33%		33.33%	33.33%	33.33%	33.33%
11/01/2020	100.00%													100.00%		100.00%					100.00%		
12/01/2020	100.00%		50.00%	50.00%	50.00%	50.00%	50.00%	50.00%				50.00%				50.00%	50.00%	100.00%	50.00%	50.00%			50.00%
02/01/2021	100.00%	100.00%										100.00%	100.00%			100.00%							

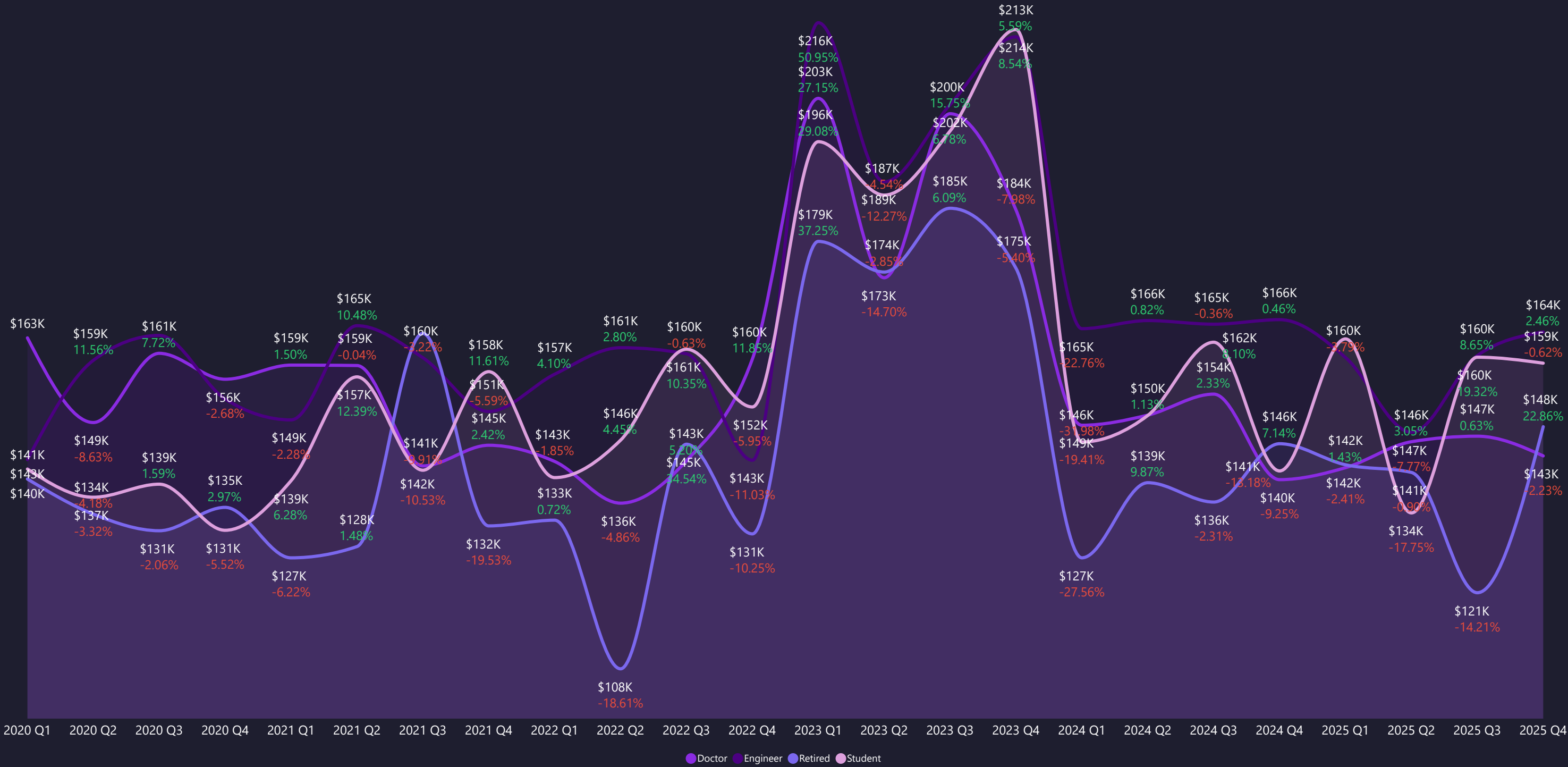


Banking Transaction



Amount Performance: QoQ Growth Analysis

% QoQ Growth





Banking Transaction

City
All

2020

2021

2022

2023

2024

2025

Avg Transaction Value

\$301.20

Purchase Frequency

16.17

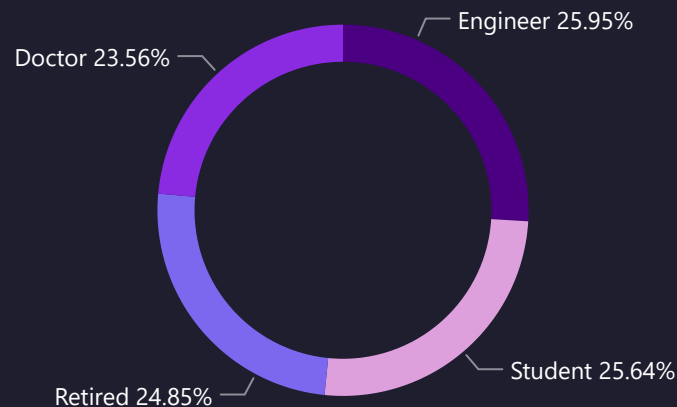
Customer Lifetime

11

Customer Lifetime Value

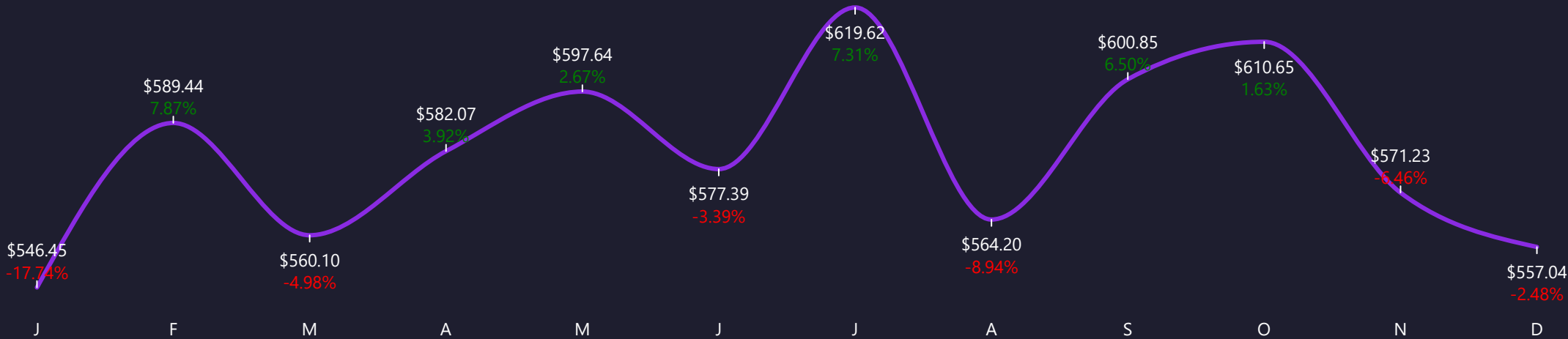
\$53.57K

CLV by Occupation Customer - 2024



Revenue per Customer by Month - 2024

% MoM Growth



AccountID	Avg Transaction Value	Total Account Balance	Total Trans Amount	ATV	Purchase Frequency	Customer Lifetime	Customer Lifetime Value
AC00363	\$441.67	\$293,708.80	\$20,317.00	\$441.67	46.00	11	\$223,487.00
AC00460	\$413.93	\$215,494.48	\$17,385.00	\$413.93	42.00	11	\$191,235.00
AC00358	\$617.16	\$124,556.60	\$15,429.00	\$617.16	25.00	11	\$169,719.00
AC00291	\$533.75	\$189,619.50	\$14,945.00	\$533.75	28.00	11	\$164,395.00
AC00208	\$738.15	\$49,028.55	\$14,763.00	\$738.15	20.00	11	\$162,393.00
AC00337	\$525.93	\$136,485.47	\$14,726.00	\$525.93	28.00	11	\$161,986.00
AC00182	\$445.58	\$108,235.57	\$14,704.00	\$445.58	33.00	11	\$161,744.00
AC00098	\$732.15	\$160,503.71	\$14,643.00	\$732.15	20.00	11	\$161,073.00
AC00304	\$358.87	\$289,615.54	\$13,996.00	\$358.87	39.00	11	\$153,956.00
AC00407	\$416.16	\$140,964.14	\$13,317.00	\$416.16	32.00	11	\$146,487.00
AC00068	\$830.25	\$44,996.74	\$13,284.00	\$830.25	16.00	11	\$146,124.00
AC00310	\$521.12	\$78,134.12	\$13,028.00	\$521.12	25.00	11	\$143,308.00



Banking Transaction

2020

2021

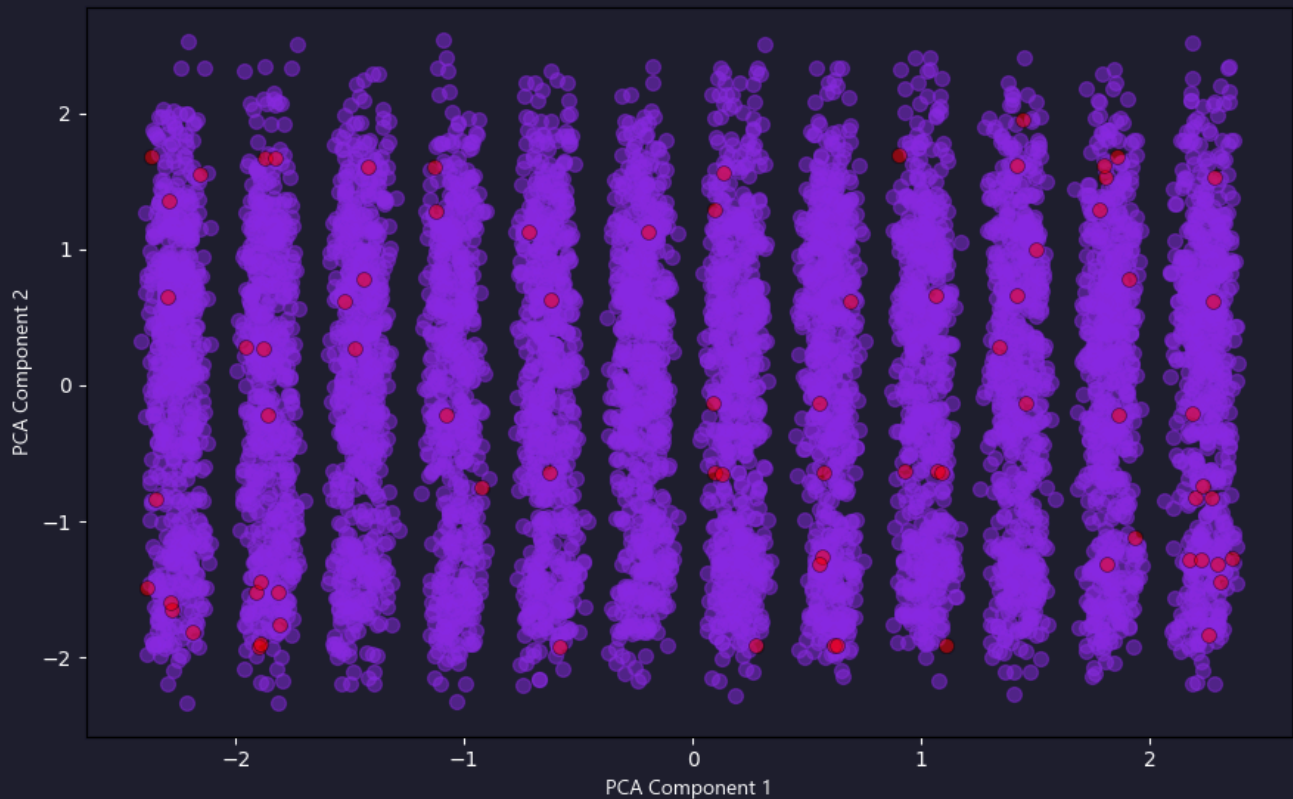
2022

2023

2024

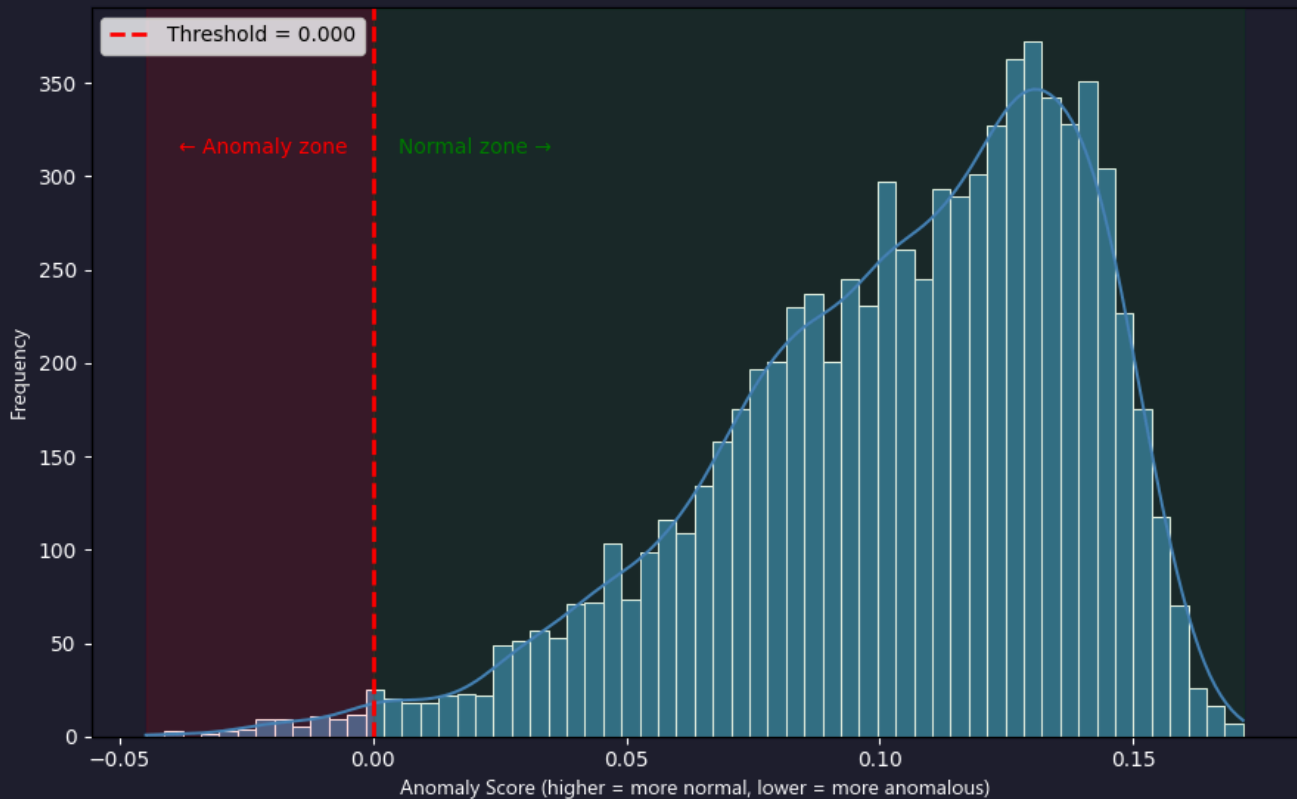
2025

Isolation Forest — Potential Frauds vs Normal (PCA 2D)



Prediction
● Normal (n=7711)
● Potential Fraud (n=78)

Distribution of Isolation Forest Anomaly Scores



TransactionID	AccountID	CustomerAge	CustomerOccupation	TransactionAmount	Location	Channel	TransactionType	TransactionDuration	AnomalyLabel	Segment Customer
TX023916	AC00071	18	Student	\$115.00	Mesa	Branch	Debit	22	Potential Fraud	Champions
TX035671	AC00071	18	Student	\$112.00	Mesa	Branch	Debit	22	Potential Fraud	Champions
TX014573	AC00317	19	Student	\$14.00	Las Vegas	Online	Debit	20	Potential Fraud	Champions
TX019000	AC00317	19	Student	\$13.00	Las Vegas	Online	Debit	20	Potential Fraud	Champions
TX028070	AC00317	19	Student	\$14.00	Las Vegas	Online	Debit	20	Potential Fraud	Champions
TX030310	AC00317	19	Student	\$13.00	Las Vegas	Online	Debit	20	Potential Fraud	Champions
TX030689	AC00353	19	Student	\$130.00	Nashville	Branch	Credit	68	Potential Fraud	Lost customers
TX031350	AC00317	19	Student	\$14.00	Las Vegas	Online	Debit	20	Potential Fraud	Champions
TX032613	AC00317	19	Student	\$14.00	Las Vegas	Online	Debit	20	Potential Fraud	Champions
TX037704	AC00317	19	Student	\$14.00	Las Vegas	Online	Debit	20	Potential Fraud	Champions



Banking Transaction

2020

2021

2022

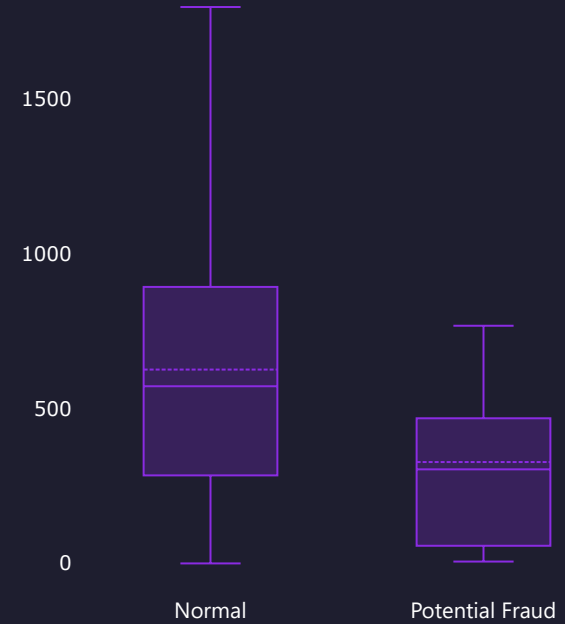
2023

2024

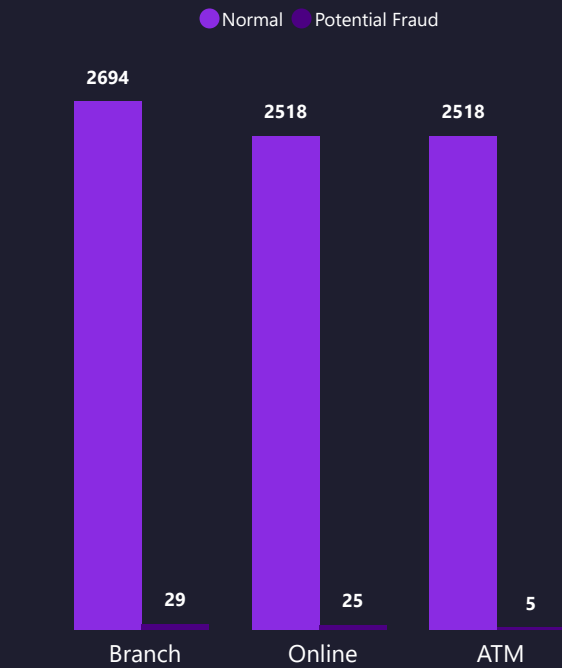
2025



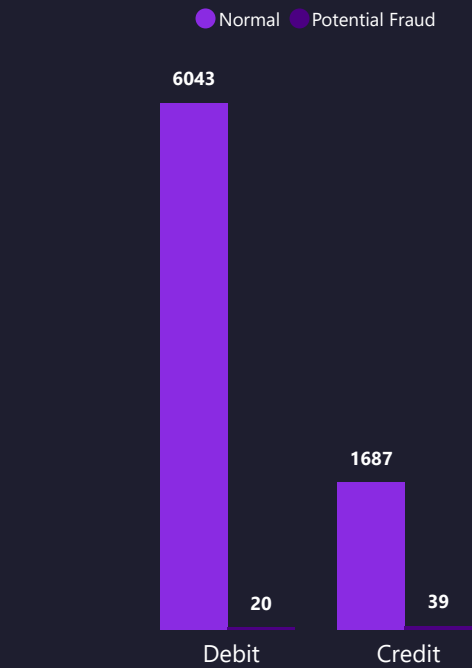
Trans Amount Distribution by Anomaly Label



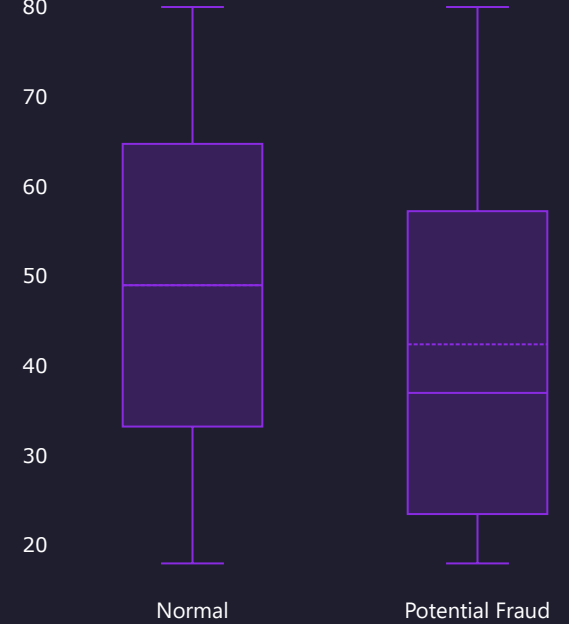
Fraud Pattern by Channel



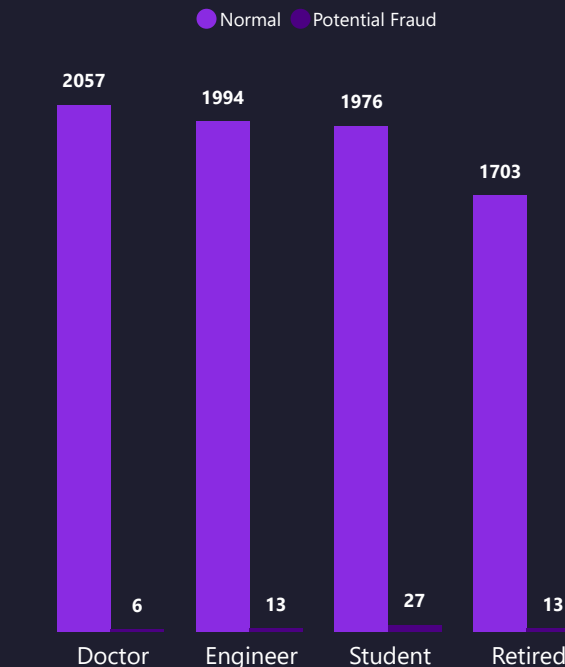
Fraud Pattern by Transaction Type



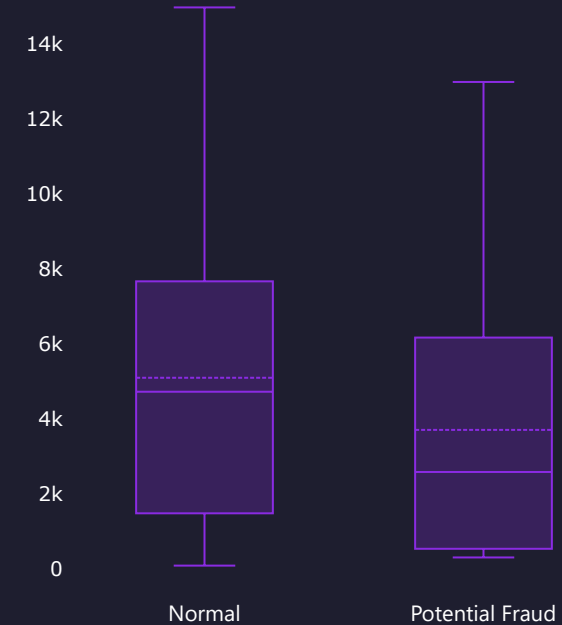
Customer Age Distribution by Anomaly Label



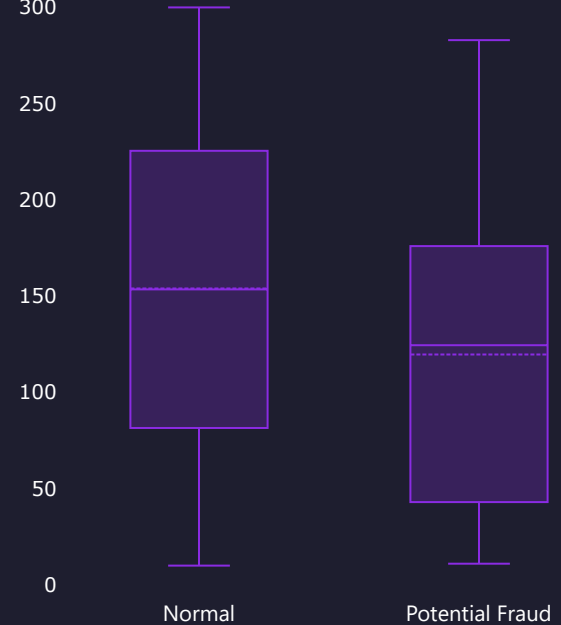
Fraud Pattern by Customer Occupation



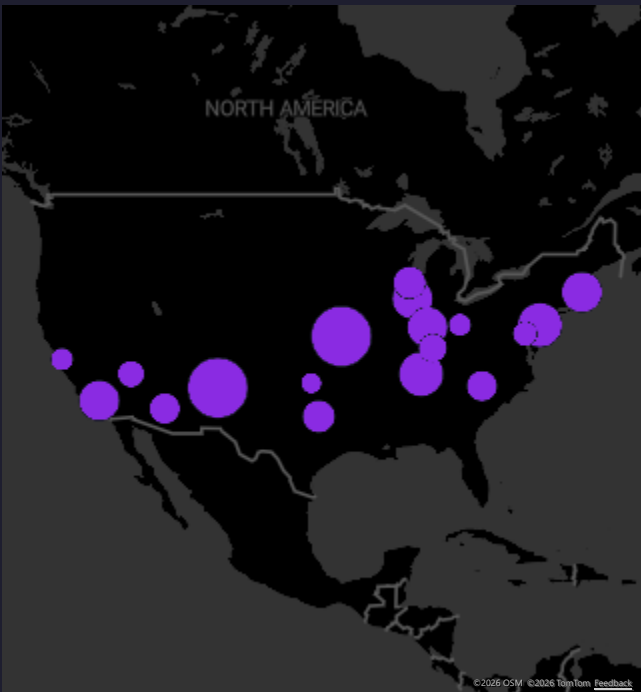
Acc Balance Distribution by Anomaly Label



Trans Duration Distribution by Anomaly Label



Geographic Distribution of Fraud





Banking Transaction

2020

2021

2022

2023

2024

2025



Monthly Trend of Avg Transaction Amount by Anomaly Label - 2022

